



BAULKHAM HILLS HIGH SCHOOL

2016
YEAR 12 TASK 3
TERM 2 ASSESSMENTS

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 60 minutes
- Write using black or blue pen
- Board-approved calculators may be used
- Show all necessary working in Questions 6-9
- Marks may be deducted for careless or badly arranged work

Total marks – 35

**Exam consists of 8 pages and
4 Questions**

Reference Sheet is on page 7 and 8.

Answer each question on the appropriate page.
All necessary working should be shown in every question.

Question 1 (10 marks)

Marks

- a) Use the substitution $u = x^2 + 1$ to find the indefinite integral

3

$$I = \int \frac{x \, dx}{\sqrt{x^2 + 1}}$$

- b) Use the substitution $x = 3 \sin \theta$ to find the integral

3

$$\int_{\sqrt{3}}^3 \sqrt{9 - x^2} \, dx$$

- c) (i) Show that $\tan^{-1} x - x^2 + \frac{\pi}{4} = 0$ has a root in the interval $1 < x < \sqrt{3}$

2

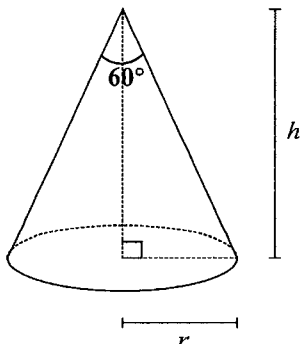
- (ii) Use one application of Newton's method to find an approximation to this root.
Take $x = 1.5$ as the first approximation expressing your answer to one decimal place.

2

End of Question 1

Question 2 (9 marks)**Marks**

- a) Grain is poured at a constant rate of 0.25 cubic metres per second. It forms a conical pile, with the angle at the apex of the cone equal to 60° . The height of the pile is h metres, and the radius of the base is r metres.



- (i) Show that $r = \frac{h}{\sqrt{3}}$ 1
- (ii) Show that V , the volume of the pile is given by $V = \frac{\pi h^3}{9}$ 1
- (iii) Hence find the rate at which the height of the pile is increasing when the height of the pile is 3 metres. 2

- b) The acceleration of the particle P is given by the equation

$$\frac{d^2x}{dt^2} = 8x - 2x^3.$$

where x metres is the displacement of P from a fixed point O after t seconds.
Initially, when $x = 2$ the velocity of the particle is 3 ms^{-1} .

- (i) Find the equation for v^2 . 3
- (ii) Between which two points the particle is moving. 2

End of Question 2

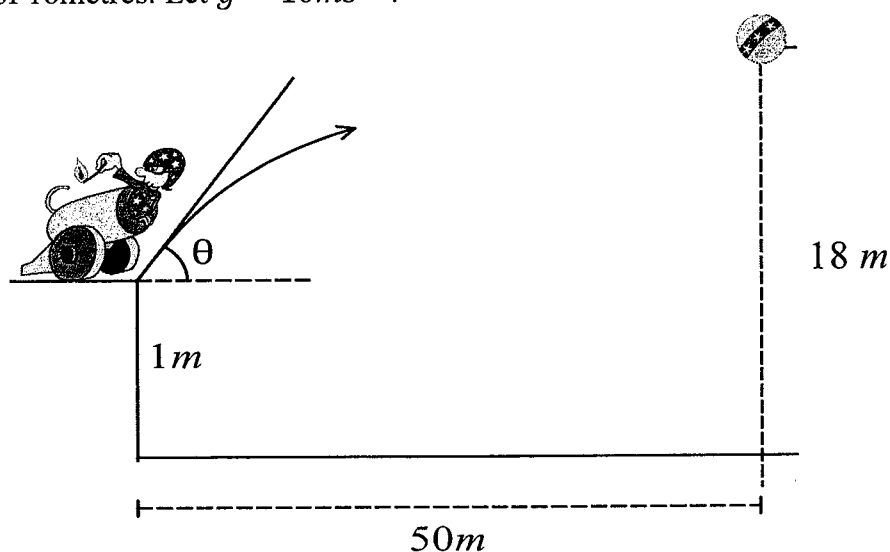
Question 3 (9 marks)

- a) A particle is moving along the x -axis, starting from a position 2 metres to the right of the origin with an initial velocity of 5 ms^{-1} and an acceleration given by $\ddot{x} = 2x^3 + 2x$.

(i) Show that $\dot{x} = x^2 + 1$. 3

(ii) Hence find an expression for t in terms of x . 2

- b) A circus act consists of a person fired from a cannon at 30 metres per second on a platform 1 metre high. The moment the cannon is fired, a ball will be released 50 metres away, from a height of 18 metres. Let $g = 10 \text{ ms}^{-2}$.



The equations of motion of the projectile of the person are :

$$x = 30t \cos \theta$$

DO NOT PROVE THESE

$$y = 1 + 30t \sin \theta - 5t^2$$

The equations of motion of the ball are:

$$x = 50$$

DO NOT PROVE THESE

$$y = 18 - 5t^2$$

- (i) What is the angle at which the cannon must be fired so the person collides with the ball? 2
- (ii) At what height does the person collides with the ball? 2

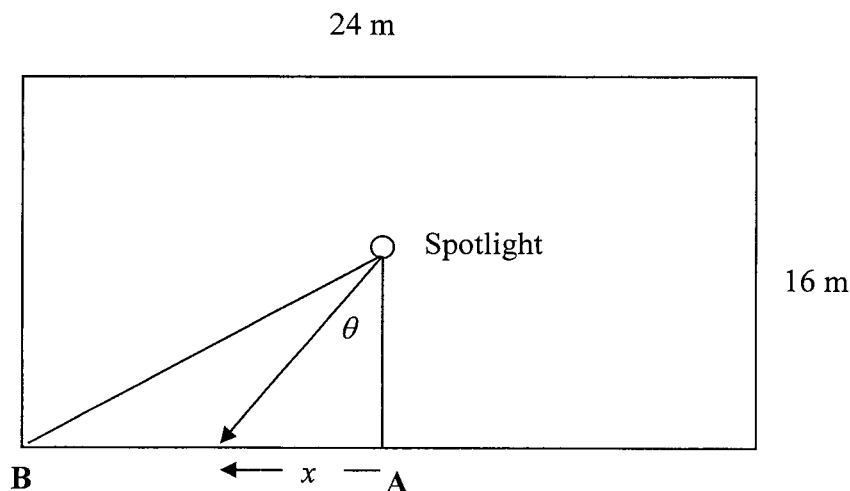
End of Question 3

Question 4 (7 marks)**Marks**

- a) A particle moving in a straight line is performing Simple Harmonic Motion. At time t seconds it has displacement x metres, where $x = 4 \cos^2 t - 2 \sin^2 t$.

- (i) Show that $x = 1 + 3 \cos 2t$. 1
- (ii) Hence express the acceleration $\ddot{x} \text{ ms}^{-1}$ of the particle in the form of $\ddot{x} = -n^2(x - b)$ 1
- (iii) Which points is the particle oscillating? 1
- (iv) At which point is the particle moving the fastest? 1

- b) A spotlight is in the centre of a rectangular nightclub which measures 24 m by 16 m. It is spinning at a rate $\frac{d\theta}{dt} = 2\pi$ radians per second. Its beam projects a spot which moves along the walls as it spins.



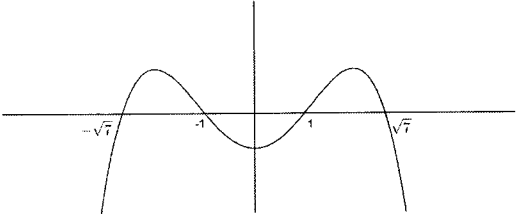
- i) Find an expression for the velocity $\frac{dx}{dt}$ in terms of x at which the spot appears to be moving along the wall from A to B. 2
- ii) What is the velocity at which the spot appears to be moving at the point A? 1

End of Exam

BAULKHAM HILLS HIGH SCHOOL
YEAR 12 EXTENSION 1 TASK 2 2016 SOLUTIONS

	Solution	Term	Marks	Comments
1a)	$u = x^2 + 1$ $du = 2x dx$ $I = \frac{1}{2} \int \frac{du}{\sqrt{u}}$ $I = \frac{1}{2} \int u^{-\frac{1}{2}} du$ $I = \frac{1}{2} \times 2 \times u^{\frac{1}{2}} + c$ $I = \sqrt{x^2 + 1} + c$		3	3 marks • Correct Solution 2 marks • Integrates correctly and leaves in terms of u • Significant progress towards the solution. 1 mark • Correct expression for integral in terms of u
1b)	$x = 3 \sin \theta$ $dx = 3 \cos \theta d\theta$	if $x = 3$, $3 = 3 \sin \theta$ $\theta = \frac{\pi}{2}$ if $x = \sqrt{3}$ $\sqrt{3} = 3 \sin \theta$ $\theta = \sin^{-1}\left(\frac{1}{\sqrt{3}}\right)$	3	3 marks • Correct Solution 2 marks • Correct integral in terms of $d\theta$ (incl. limits) 1 mark • Correct integral in terms of $d\theta$
	$I = \int_{\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)}^{\frac{\pi}{2}} \sqrt{9 - 9 \sin^2 \theta} \times 3 \cos \theta d\theta$ $I = \int_{\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)}^{\frac{\pi}{2}} 3 \cos \theta \times 3 \cos \theta d\theta$ $I = 9 \int_{\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)}^{\frac{\pi}{2}} \frac{1}{2} (\cos 2\theta + 1) d\theta$ $I = \frac{9}{2} \left[\frac{1}{2} \sin 2\theta + \theta \right]_{\sin^{-1}\left(\frac{1}{\sqrt{3}}\right)}^{\frac{\pi}{2}}$ $I = \frac{9}{2} \left[\left(\frac{1}{2} \sin \pi + \frac{\pi}{2} \right) - \left(\frac{1}{2} \sin \left(2 \sin^{-1} \left(\frac{1}{\sqrt{3}} \right) \right) + \frac{\pi}{6} \right) \right]$ $I = 2.18 (2dp)$			
1(i)	Let $f(x) = \tan^{-1} x - x^2 + \frac{\pi}{4}$ $f(1) = \tan^{-1} 1 - 1^2 + \frac{\pi}{4}$ $f(1) = 0.57 > 0$ $f(\sqrt{3}) = \tan^{-1} \sqrt{3} - \sqrt{3}^2 + \frac{\pi}{4}$ $f(\sqrt{3}) = -1.16 < 0$ Since $f(1) > 0$, $f(\sqrt{3}) < 0$ and $f(x)$ is continuous, then there is at least one root between $x = 1$ and $x = \sqrt{3}$		2	2 marks • Correct Solution 1 mark • States $f(x)$ is continuous • shows $f(1) > 0$ and $f(\sqrt{3}) < 0$

	Solution	Marks	Comments
1c(ii)	$f'(x) = \frac{1}{x^2 + 1} - 2x$ $f'(1.5) = \frac{1}{1.5^2 + 1} - 2(1.5) = -2.692$ $x_1 = x_0 - \frac{f(1.5)}{f'(1.5)}$ $x_1 = 1 - \frac{0.4818}{-2.692} = 1.1127 \dots$ $x_1 = 1.3$	2	2 marks • Correct Solution 1 mark • Correct use of formula with wrong values • finding $f'(1.5)$
2a(i)	In $\triangle OPA$, $\angle OPA = \frac{1}{2} \times 60^\circ = 30^\circ$ $\tan 30^\circ = \frac{r}{h}$ $r = h \tan 30^\circ$ $r = \frac{h}{\sqrt{3}}$	1	1 mark • Correct Solution
2a(ii)	Volume of the cone $= \frac{1}{3} \pi r^2 h$ $V = \frac{1}{3} \pi \left(\frac{h}{\sqrt{3}} \right)^2 h$ $V = \frac{\pi h^3}{9}$	1	1 mark • Correct Solution
2a(iii)	$\frac{dV}{dt} = 0.25$ $\frac{dV}{dh} = \frac{\pi h^2}{3}$ When $h = 3$, $\frac{dV}{dh} = 3\pi$ $\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$ $0.25 = 3\pi \times \frac{dh}{dt}$ $\therefore \frac{dh}{dt} = \frac{1}{12\pi}$ $\therefore$ the height of the pile is increasing at a rate of $\frac{1}{12\pi} \text{ ms}^{-1}$ when $h = 3$	2	2 marks • Correct Solution 1 mark • Finds a correct expression using $\frac{dV}{dt} = \frac{dV}{dh} \times \frac{dh}{dt}$
2b(i)	$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 8x - 2x^3$ $\frac{1}{2} v^2 = 4x^2 - \frac{1}{2} x^4 + c$ $v^2 = 8x^2 - x^4 + 2c$ When $v = 3, x = 2$ $3^2 = 8(2)^2 - (2)^4 + 2c$ $7 = 2c$ $v^2 = 8x^2 - x^4 + 7$	2	2 marks • Correct Solution 1 mark • Uses $\ddot{x} = \frac{d}{dx} \left(\frac{1}{2} v^2 \right)$

	Solution	Marks	Comments
2b(ii)	$v^2 = (7 - x^2)(1 - x^2)$ $v^2 = (\sqrt{7} - x)(\sqrt{7} + x)(1 - x)(1 + x)$ Since $v^2 \geq 0$  $-\sqrt{7} \leq x \leq -1$ or $1 \leq x \leq \sqrt{7}$ Since it initially starts at $x = 2$ $\therefore 1 \leq x \leq \sqrt{7}$ (Note A better students solution would have explained: that at $x = 2, v = 3$. As $x = \sqrt{7}, a < 0$ (ie. gradient of $v < 0$) so the particle moves left. As $x = 1, a > 0$ (ie. gradient of $v > 0$) so the particle moves right. And the particle will oscillate between $1 \leq x \leq \sqrt{7}$ and the motion is not in SHM.)	2	<u>2 marks</u> • Correct Solution <u>1 mark</u> • Finds all values of $v = 0$
3a(i)	$\frac{d}{dx} \left(\frac{1}{2} v^2 \right) = 2x^3 + 2x$ $\frac{1}{2} v^2 = \frac{1}{2} x^4 + x^2 + c$ When $t = 0, x = 2, v = 5$ $\frac{1}{2} (5)^2 = \frac{1}{2} (2)^4 + (2)^2 + c$ $c = \frac{1}{2}$ $v^2 = x^4 + 2x^2 + 1$ $v = \pm \sqrt{(x^2 + 1)^2}$ When $x = 2, v = 5$ $v = x^2 + 1$	3	<u>3 marks</u> • Correct solution by considering $\pm$ cases <u>2 marks</u> • Finding a correct expression for v <u>1 mark</u> • Finding c
3a(ii)	$\frac{dx}{dt} = x^2 + 1$ $\frac{dt}{dx} = \frac{1}{x^2 + 1}$ $t = \tan^{-1} x + c$ When $t = 0, x = 2$ $c = -\tan^{-1} 2$ $t = \tan^{-1} x - \tan^{-1} 2$	2	<u>2 marks</u> • Correct Solution <u>1 mark</u> • Correct expression for $\frac{dt}{dx}$

	Solution	Marks	Comments
3b(i)	$x = 30t \cos \theta$ $y = 1 + 30t \sin \theta - 5t^2$ $x = 20$ $y = 18 - 5t^2$ To intercept, the x and y values must be equal. $30t \cos \theta = 20$ $t \cos \theta = \frac{5}{3} \quad \dots (1)$ $1 + 30t \sin \theta - 5t^2 = 18 - 5t^2$ $t \sin \theta = \frac{17}{30} \quad \dots (2)$ $\frac{(1)}{(2)} = \frac{t \sin \theta}{t \cos \theta} = \frac{5}{3} \div \frac{17}{30}$ $\tan \theta = \frac{17}{50}$ $\theta = 18^\circ 47' \quad (\text{to the nearest minute})$	2	<u>2 marks</u> • Correct Solution <u>1 mark</u> • Equates the x and y equations • Equates one equation to find θ
3b(ii)	$t = \frac{17}{30 \sin 18^\circ 47'}$ $t = 1.76 \text{ sec}$ $y = 18 - 5(1.76)^2$ $y = 2.51 \text{ m}$	2	<u>2 marks</u> • Correct Solution <u>1 marks</u> • Finds t
4a(i)	$x = 1 + 3 \cos 2t$ $x = \cos^2 x + \sin^2 x + 3(\cos^2 t - \sin^2 t)$ $x = 4 \cos^2 x - 2 \sin^2 t$	1	<u>1 marks</u> • Correct Solution
4a(ii)	$\dot{x} = -6 \sin 2t$ $\ddot{x} = -12 \cos 2t$ $\ddot{x} = -4(3 \cos 2t)$ $\ddot{x} = -4(x - 1)$ Where $n = 2$ and $b = 1$	1	<u>1 marks</u> • Correct Solution
4a(iii)	the particle oscillates between $x = -2$ and $x = 4$ $\text{ie. } -2 \leq x \leq 4$	1	<u>1 marks</u> • Correct Solution
4a(iv)	$x = 1 + 3 \cos 2t$ The particle moves fastest at the centre of motion at $x = 1$	1	<u>1 mark</u> • Correct Solution

	Solution	Marks	Comments
4b(i)	$\tan \theta = \frac{x}{8}$ $x = 8 \tan \theta$ $\frac{dx}{d\theta} = 8 \sec^2 \theta$ $\frac{dx}{dt} = \frac{dx}{d\theta} \cdot \frac{d\theta}{dt}$ $= 8 \sec^2 \theta \times 2\pi$ $= 16\pi \sec^2 \theta$ $= 16\pi (1 + \tan^2 \theta)$ $= 16\pi (1 + \tan^2 \theta)$ $= 16\pi \left(1 + \left(\frac{x}{8} \right)^2 \right)$ $= 16\pi + \frac{\pi^2 x}{4}$	2	<u>2 marks</u> • Correct Solution <u>1 mark</u> • Finds $\frac{dx}{d\theta}$ • Correct use of the chain rule with substitution.
4b(ii)	At A, $x = 0$ $\frac{dx}{dt} = 16\pi \left(1 + \left(\frac{0}{8} \right)^2 \right)$ $= 16\pi$	1	<u>1 mark</u> • Correct Solution